



Lucas Furer

Computer Science graduate

A detail-oriented and enthusiastic MSc Computer Science graduate with a passion for physics-based simulations, computer graphics and high-performance computing. I am an eager learner who enjoys collaborating with others and exploring new technologies. Experienced with C++, Java, HLSL and python.

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 Delft, NL

Education

- M.Sc. Computer Science, TU Delft** **2023-2026**
- **Thesis: FMM-t-SNE** Implemented a fast multipole N-body solver and integrated it into t-SNE. Developed in C++ with OpenGL visualization
 - **Grade:** 8.5
 - **Notable courses:** Adv. Digital Image Processing, Category Theory, Geometry Processing
- B.Sc. Computer Science, TU Delft** **2019-2023**
- **Thesis: Soft Peaks** Studied how the distribution of ray reflections evolve as smoothing is applied to microstructures. Implemented in C++ with results rendered to .ppm
 - **Grade:** 7.5
 - **Minor:** Security, Safety & Justice
 - **Notable courses:** Intro to Quantum Computing, Big Data Processing, Computer Graphics

Experience

Internship

- Feedback Analytics, Software Engineer** **2022**
- Developed a visualization tool for user-created questionnaires using Angular 13
 - Lead a five-person team, collaborating closely with the client to gather requirements

Projects

- Independent Game Development Project, Software Engineer** **2025**
- Organized a four-person team to create a game using Unity and C#
 - Developed procedural terrain generation using Perlin noise and the Marching Cubes algorithm
- Blender plugin project, CS4540 Geometric Data Processing university project** **2024**
- Developed Blender plugins in python using algorithms such as Constrained Laplace Smoothing, Point-to-Point ICP, Point-to-Plane ICP and KD-tree nearest-neighbor search
 - Conducted experimental performance analysis, comparing convergence speed, registration accuracy and algorithm trade-offs across multiple 3D datasets.

Skills

Programming

C++
Libraries: Boost, Eigen, Fastor, GLM, OpenGL, GLFW, FFTW, Vulkan

C, Python, C#, Java, HLSL, CUDA, Unity
Libraries: Numpy, Matplotlib, Pandas, OpenTSNE, Sklearn

x86-Assembly, HTML, CSS, Javascript, Scala, Rocq, Haskell, SQL, MiniZinc

Tools

CMake, Docker, Linux, Bash, Git, Blender, Markdown, LaTeX

Language

Dutch, Native English, Fluent

Extracurricular Activities

- D.S.M.G. Krashna Musika, Bass singer** **2024-2026**
- Performed with a classical music student association in concerts across Europe
 - Developed teamwork, discipline and performance skills through rehearsals and live performances
- D.S.B.V. Yoroshi** **2024-2025**
- Practised judo at a student budosport association

Interests

Playing piano, creating games/simulations, Running, bouldering